

AKADEMIKA

AL-DL

**FIBER OPTIC ANALOG & DIGITAL LIN
A660nm FOR PLASTIC FIBER**

**EXPERIMENTAL & SERVICE
MANUAL**

.... A MESSAGE FROM

Today's system designers are faced with tomorrow's problems. FIBER OPTIC COMMUNICATION is one of the important subject need to teach while learning electronics.

It is our vision to provide you with the product you need for training ensuring lasting reliability & quality.

OUR MOTTO;

- Light years ahead – refers to leadership.

As leaders in our industry in India, We are totally committed to servicing as the standard against which all are measured in the areas of

• Design • Quality • Value • Delivery • Support.

We are truly light years ahead of our competition in this area. That means that you our valued customers are guaranteed satisfaction.

As you will move through this manual you will quickly discover that We have a complete, truly innovative & superior training products we are so committed to quality that we back our products with a complete comprehensive warranty.

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SAFETY RULES

Carefully follow the instructions contained in this handbook, as they provide you with important points on safety during the installation, use and maintenance. Keep this handbook at hand for any further help.

Set all accessories in order on unpacking, and check integrity with respect to its checklist. Check the Kit for any visible damage.

Before connecting the power supply to the equipment, be sure patch cords are connected correctly, jumpers are correct as per experimental position.

In **AL-DL kit**, signals are to be monitored with an oscilloscope, the scope input channel should be a.c. coupled, unless otherwise indicated. Ensure that X10 oscilloscope probes are used throughout.

These kits/System must be employed only for the use for which it has been conceived, i.e. as educational kits, and must be used under the direct survey of expert personnel. Any other use is improper and so dangerous. The manufacturer cannot be considered responsible for eventual damages due to improper, wrong or unreasonable uses.

In case of fault or malfunctioning of the trainers, turn off the power supply and do not tamper the kits. In case servicing is required, contact the service center for technical assistance.

The kits are liable to malfunction/under-perform if they are not operated under standard environmental conditions of temperature and humidity.

WARRANTY

These kits are warranted against defects in workmanship and materials. If any failure resulting from a defect in either workmanship or material should occur under normal use within a year from the original date of purchase, such failure will be corrected free of charge to the purchaser by repair or replacement of defective part or parts. When the failure is result of user's neglect, natural disaster or accident, we charge for repairs regardless of the warranty period. The warranty does not cover perishable items like connecting patch chords, Crystals, etc. and other imported items.

This warranty is subject to the following conditions and limitations. The warranty is void and inapplicable if the defective product is not brought or sent to our authorized service center or sales outlet within the warranty period. Defective product is, on Akademika Lab Solution sole judgment. The defective product will be replaced with new one or repaired without charge or with charge.

In the event warranty service is needed, purchaser should get in touch with service center or sales outlet. The purchaser should return the product to the service center or sales outlet at his or her sole expense. The loss and damage in transit will be outside the preview of this warranty. A returned product must be accompanied by a written description of the defects. Type and Model No. of kit /system has to be mentioned specifically. We return the product to the purchaser at our expense. In case that warranty does not cover the product on Akademika Lab Solution judgment, we repair the product after obtaining prior permission from purchaser who receives an estimate statement about repairing charges. In such case, Akademika Lab Solution bares the transporting expenses required to send back all the repaired products for the moment, and then repairs and transporting expenses will be charged against the purchaser by the statement of accounts.

When the authorized sales agents sell our products, they must notify the purchaser of the warranty contents, but have no rights to stretch the meaning of original warranty contents or offer additional warranty. Akademika Lab Solution does not provide any other promise or suggestive warranty and hold no liability for the damage caused by negligence, abnormal use or natural disaster. Akademika Lab Solution is not responsible for the damages even if it is notified of above dangers in advance as well.

For more special service or overall repairs and maintenance of old and decrepit products, be sure to contact our service center or sales outlet.

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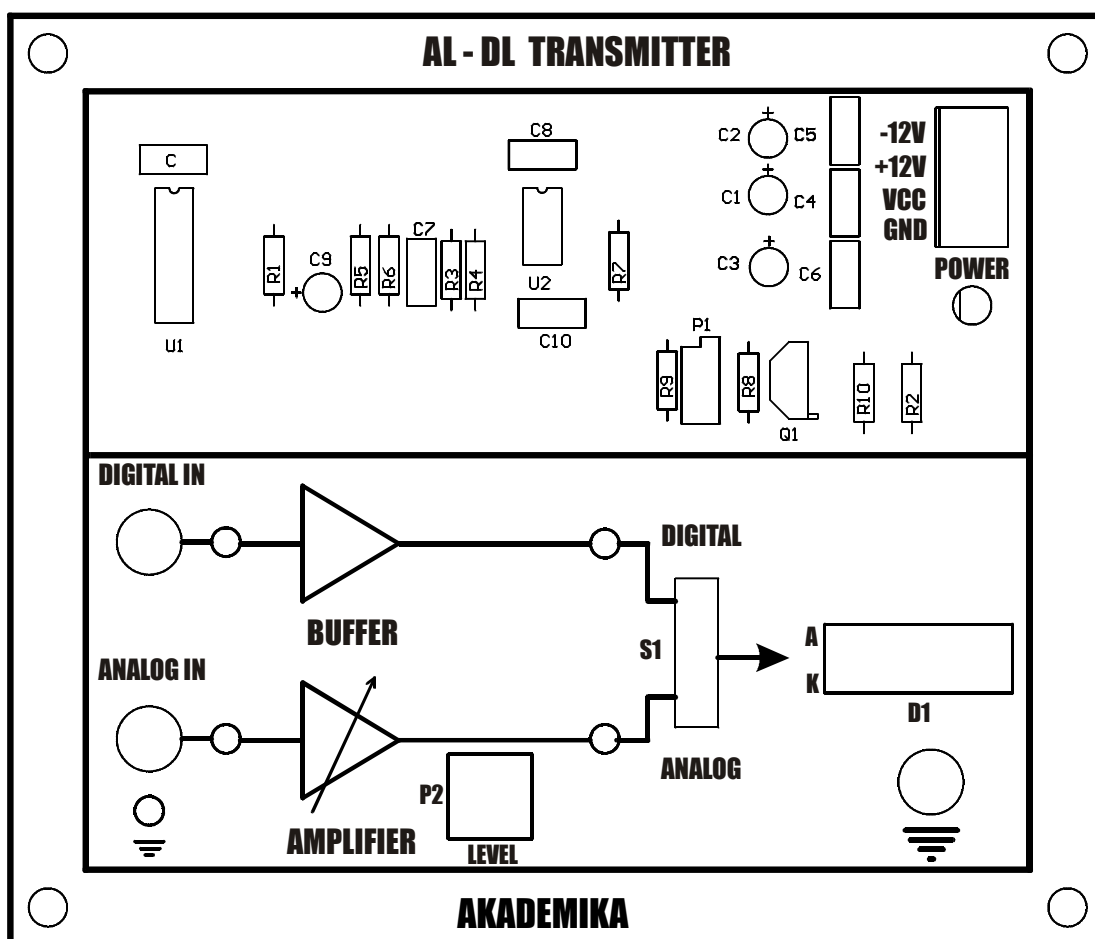
TECHNICAL SPECIFICATIONS

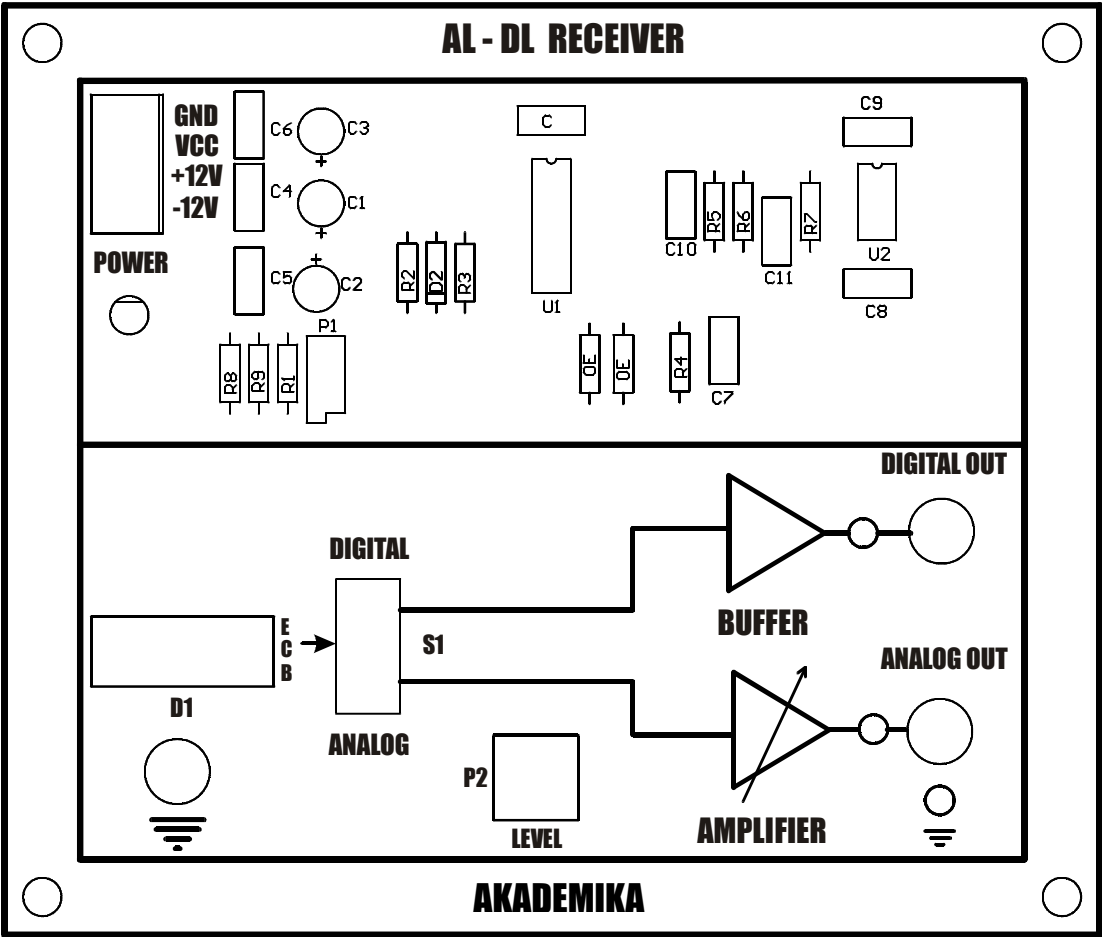
Transmitter	: 1 No Glass Fiber Optic LED : Peak Wavelength of Emission 660nm Red Visible (SFH750V)
Receiver	: 1 NO. Siemens Fiber Optic Photo Detector : Photo Transistor Detector With (SFH 350V)
Bandwidth	: 200 Khz
Fiber Optic Cable	
Plastic fiber	
Type	: Step Index, Multimode
Length	: 1 Meter, 3 Meter
Numerical Aperture	: 0.5
Interconnection	: 2mm Banana socket
Power Supply	: GND, +5V, +12V,-12V

Deliverables

Mini Links	: 04 Nos.	
Plastic Fiber Step Index, Multimode, 1000um, 1mtr	: 01 No	
Plastic Fiber Step Index, Multimode, 1000um, 3M	: 01 No	
Steel Rule	: 01 No	
ACR-14/0 N.A Jig	: 01 No	
AL-DL EXPT& SER Manual	: 01 No	:
Power Supply	: 01 No.	
Power Chord	: 01 No.	

FUNCTIONAL BLOCK





INTRODUCTION

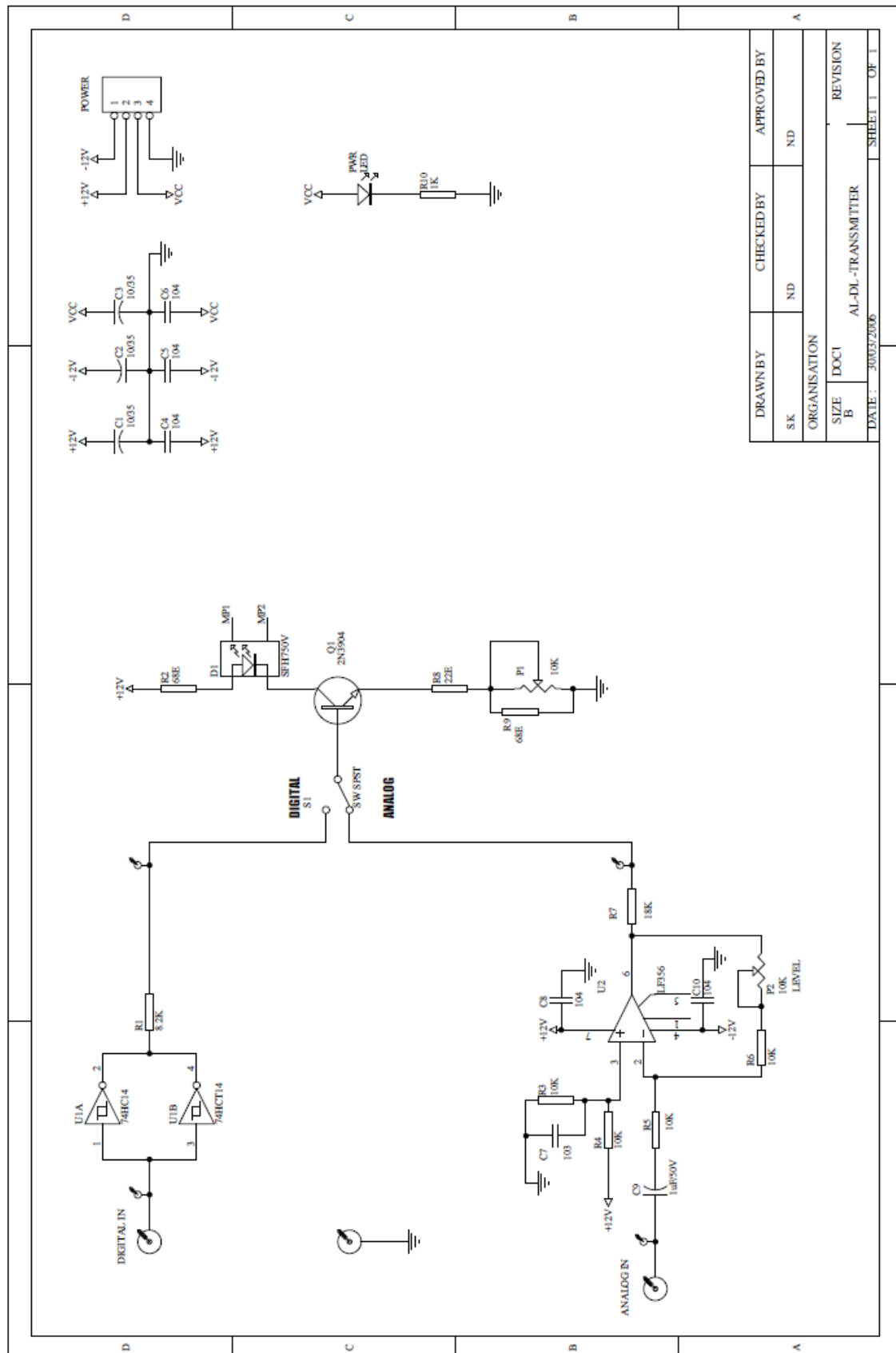
Fiber Optics technology is rapidly becoming a familiar & indispensable part of human life. You have undoubtedly heard of Fiber Optics. All giant telephone companies like AT & T, sprints have saturated the airways & print media with advertisements heralding this bright & new technology. Futurists, science writers & entrepreneurs forecast fantastic growth for Fiber Optics applications.

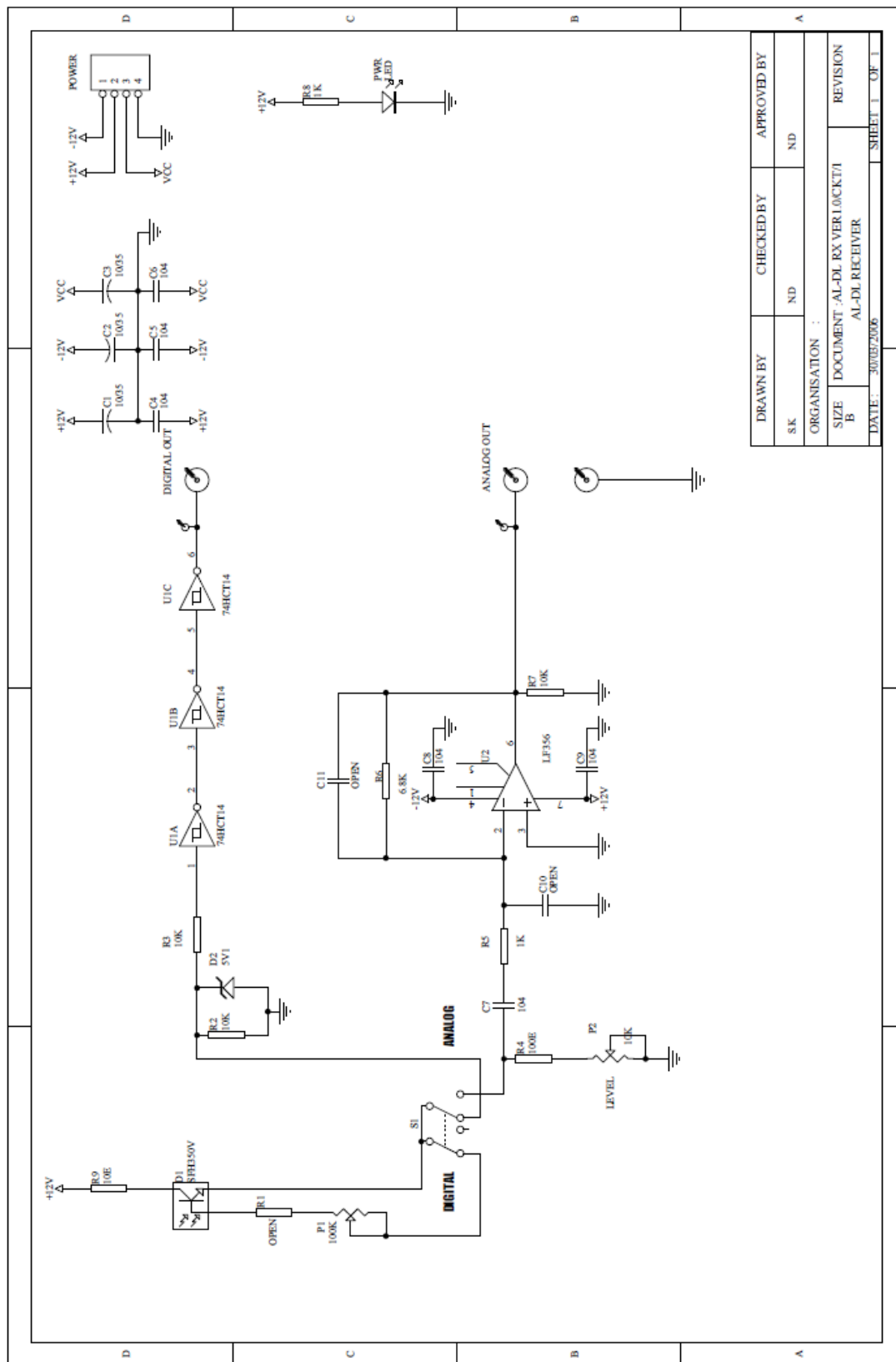
And their enthusiasm is valid. Today medical fiber optic systems allow physicians to peer inside the human body without surgery. Military commanders demand portable battlefield communication systems, which use superior fiber optic transmissions. The massive bundles of copper wiring which once carried telephone conversations between continents now are being replaced with the optical fibers from coast to coast.

Very few technologists ever come close to the fantastic growth rate that market analysts love for them to predict. Fiber optics has exceeded those rates. The next decade promises even more wonders. The influence of fiber optics will pervade our homes, workplaces & recreational facilities. Your television reception will take on startling clarity. Newsprint may largely become a thing of the past as environmentally friendly electronics newspapers become feasible. Two-way fiber optic will allow you to “attend college” in the comfort of your home.

Anyone who takes an interest in fiber optics & pursues a career in the field today is on the ground floor of opportunity. The technology is sufficiently new that few “experts” exist. Many science colleges, universities & engineering colleges have included fiber optics in their syllabus. The future beckons exciting with the prospect of making new discoveries finding new applications for the technology.

CIRCUIT DIAGRAM





ABOUT FIBER OPTICS

Before fiber optics came along, the primary means of real time data communication was electrical in nature. It was accomplished using copper wire or by transmitting electromagnetic (radio) waves through free space. Fiber optics changed that by providing an alternative means of sending information over significant distances – using light energy. Although initially a very controversial technology, today fiber optics has been shown to be very reliable & cost effective. Light as utilized for communication has a major advantages because it can be manipulated (modulated) at significant higher frequencies than electrical signals can. For example, a fiber optic cable can carry up to 100 million times more information than a telephone line! The fiber optic cable has lower energy loss & wider bandwidth capabilities than copper wire.

As you will learn, fiber optic communication is a quite simple technology, closely related to electronics. In fact, it was research in electronics that set the stage for fiber optics to develop into the communication giant that it is today. Fiber optics became reality when several technologies came together at once. It was not an immediate process, nor was it easy, but it was most impressive when it occurred. An example of one critical product, which emerged from that technological merger, was the semiconductor LED that we used in our kit.

How Optical Fiber Works?

The principle of operation of optical fiber lies in the behavior of light. Light travels in straight line through most optical materials, but that's not necessarily the case at the junction (interface) of two materials of different refractive indices. Air & water are a case in point as shown in FIG. A. The light ray traveling through air actually is bent as it enters the water. The amount of bending depends on the refractive indices of the two materials involved & also on the angle of the incoming (incident) ray of light as it strikes the interface. The angle of the incident ray is measured from a line drawn perpendicular to the surface. The same is true for the angle of the outgoing (transmitted) ray of light after it has been bent.

The mathematical relationship between the incident ray & the refracted ray is explained by Snell's Law

$$n_1 \bullet \sin \Theta_1 = n_2 \bullet \sin \Theta_2$$

In which n_1 & n_2 are the refractive indices of the initial and secondary materials respectively and Θ_1 and Θ_2 are the incident and transmitted angles.

Snell's Law says that refraction (bending) of light cannot take place when the angle of incidence grows too large (as when light travels from a material with a high refractive index to one with a low refractive index). If the angle of incidence exceeds a certain critical value (in which the product on n_1 & the sine of the angle equals or exceeds one), light cannot exit

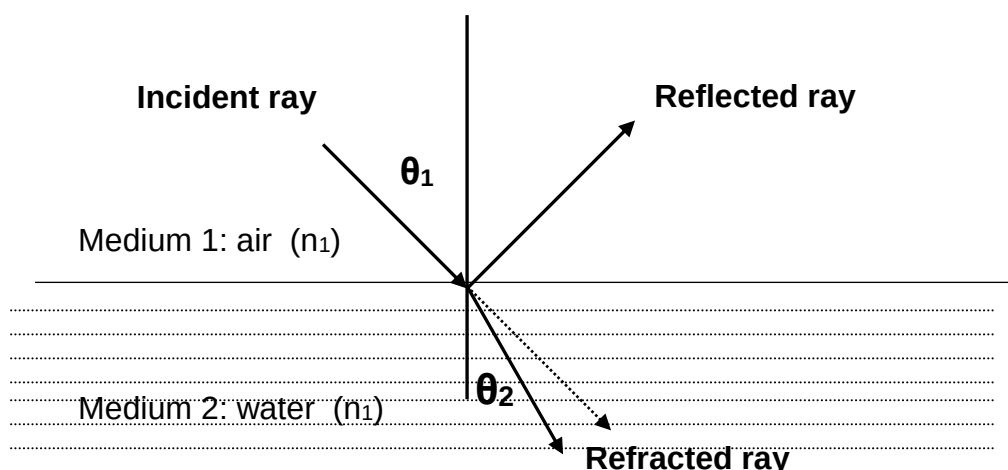


Fig. A: The Different Portions Of A Light Ray At A Material Interface

If light cannot exit the material, it is reflected. The angle that is reflected is equal to the angle of incidence. The phenomenon just described is called Total Internal Reflection and it is what keeps light inside an optical fiber.

Types of Optical Fiber

The simplest fiber optic cable consists of two concentric layers of transparent materials. The inner portion (the core) transports the light. The outer covering (the cladding) must have a lower refractive index than the core, so the two are made of different materials. The cable used in this kit also has a jacket to protect the optical properties of the core & cladding.

Optical fiber is generally made from either plastic or glass. This kit uses plastic fiber, which is very easy to terminate, & does not require special tools. Plastic fiber is generally limited to uses involving distance of less than 100 meters because of high attenuation (higher loss).

Glass fiber on other hand has very low attenuation (low loss) & hard to cut, requires special end connections & is more expensive.

The core of fiber is typically made of silica doped with impurities, which increases the refractive index relative to pure silica. The cladding, which surrounds the fiber core & creates an optical interface, is typically made from pure silica. The outer buffer coating is a plastic cover.

The first type of fiber optic cable put to use was called Step Index. In this design the cladding has a different index of refraction from the core. The light bounces off the sides & is reflected back into the fiber core. The problem with this design is that the reflected light must travel a slightly longer distances than that which travels down the center of the fiber, thus limiting the max. transmission rate. This design was improved with the use of Graded Index Fiber. In this design, the index of refraction decreases in proportion to the distance away from the center of the fiber core. The light moves more quickly in the outer portion, thus compensating for the additional distance. The change in index also has the effect of 'bending'. The light reflects back towards the core. This change increases the transmission capacity. In the newest Single Mode design, the diameter of the fiber core is made so small that all the light travels in a straight line. Even the latest fiber optic facility in use today uses less than 5% of the max. Theoretical capacity of a Single mode Fiber

Single Mode V/S Multimode

There are essentially two different types of fiber optic transmission schemes in use. Multimode fiber optic systems are typically used in short haul applications such as LANS & FDDI. Recent growth in multimode installations has been driven by LAN/FDDI applications. The term Multimode means that the diameter of the fiber optic core is large enough to propagate more than one mode (electromagnetic wave). Because of the multiple modes the pulse that is transmitted down the fiber tends to become stretched over distance. This is referred to as Modal Dispersion & has the effect of reducing available bandwidth.



Fig. B: Modal Dispersion Of Light

Typical wavelengths used in multimode applications are 850 nm & 1300 nm. There are other wavelengths used (e.g. 660 nm for plastic fiber, 820/870 nm) in specialized applications such as the military. The vast majority of multimode applications are 850 or 1300 nm wavelengths. Most multimode cables in use today are graded index in type with a cladding diameter of 125 microns & fiber core diameter of 62.5 or 50 microns (Typically referred to as 62.5/125 or 50/125 fiber)

Single Mode Systems are generally used in long haul applications (10's or 100's of kilometers) & local loop applications in both the carrier & CATV markets. Some large utilities such as electrical power companies also utilize single mode fiber optic backbones. As the name implies single mode fiber is designed to propagate only one mode of light. It is therefore not affected by modal dispersion & has higher bandwidth capacity. It is considered to be higher quality transmission system. Wavelengths used in single mode applications are 1300 & 1550 nm. The majority of today's installed systems uses 1300 nm wavelength technology & is designed for higher speed applications. These systems are more sensitive to back reflections from connectors & sharp cable bends. They often require additional tests, such as Optical Return Loss, to be run. Most single mode cable in used today are step index in type with cladding diameter of 125 micron & fiber core diameter of 9 micron (Typically referred to as 9/125 fiber).

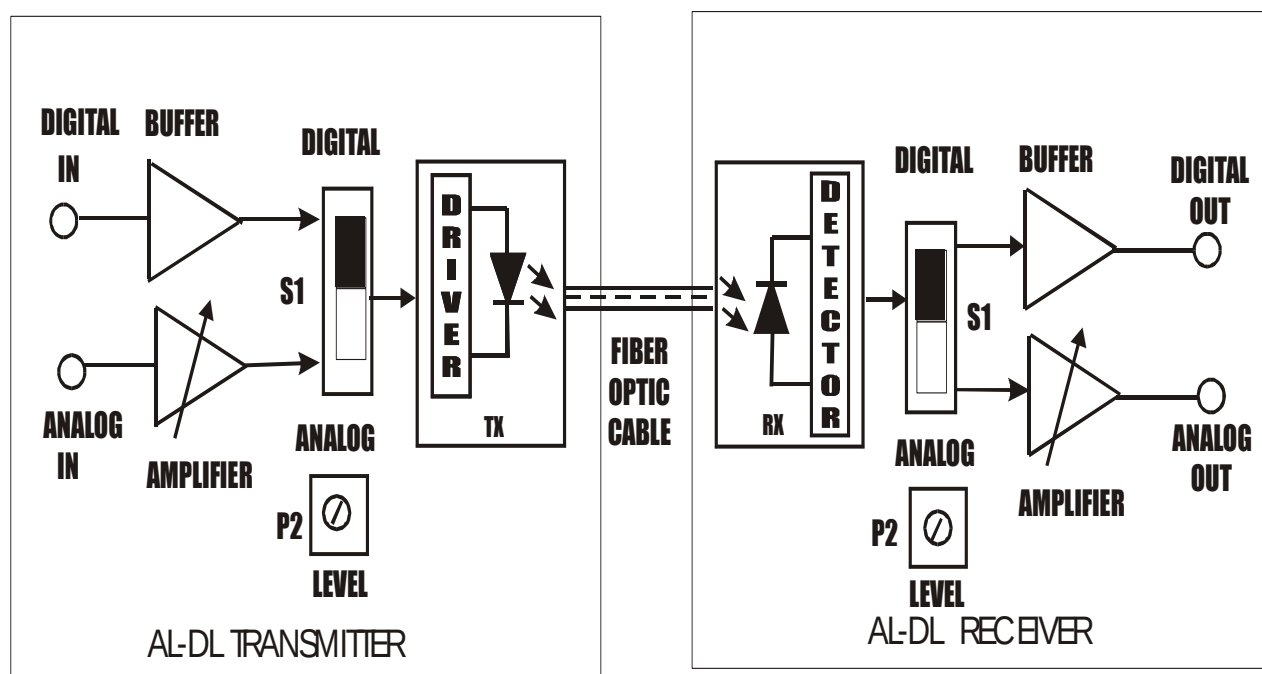
Advantages Of Fiber Optics Over Conventional Copper Cables

- Much greater bandwidth (Several orders of magnitude are theoretically possible).
- Immunity to electrical disturbances, ground loops, cross talks etc. in addition, no EMI radiation is generated.
- Much lighter. A 3-inch bundle of 900 twisted copper pairs can carry about 21,000 channels of traffic & weighs about 25,000 pounds/mile. A 1/2 inch fiber link with 12 fiber strands can carry about 3,00,000 channels & weighs about 200 pounds/mile. In addition less duct space is required to make the cable within building.

-
- Better in hostile environment, not affected as much by temperature, water etc.
 - Lower transmission loss. Typical loss on a fiber link is 0.2 dB/Km; on a copper based facility, one can usually expect at least 5 db/km. For longer links, fewer repeaters are required.
 - Better security as it is not possible to simply bridge onto the facility & monitor the traffic.

With all these properties of optical fiber, it is becoming very important in sensing field also. Most of the properties of optical fiber are useful for sensors. Further optical fiber sensors offer extreme sensitivity & can be used to sense almost any physical parameters like temperature, velocity, current, voltage etc.

EXPERIMENT NO.1



EXPERIMENT NO.1

NAME

Setting Up a Fiber Optic Digital Link

OBJECTIVE

The objective of this experiment is to study a 660nm Fiber Optic Digital Link. In this experiment, you will study the relationship between the input signal & received signal.

THEORY

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

Transmitter

Fiber optic transmitters are typically composed of a buffer, driver & optical source. The buffer electronics provides both an electrical connection & isolation between the transmitter & the electrical system supplying the data. The driver electronics provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern. The LED SFH750V supplied with the link operates within the visible light spectrum. Its optical output is centered at wavelength of 660 nm.

The LED **SFH750V** used in the link is coupled to the transistor driver in common emitter mode. The driver is preceded by the buffer. The buffer in this case is a 74HC04-inverting gate configured as voltage follower. In the absence of input signal no voltage appears at the base of the transistor. This biases the transistor to the cutoff region for linear applications. Thus LED emits no intensity of light at this time. When the signal is applied to the input post it biases the transistor to the active region for linear applications. Thus LED emits full intensity of light at this time. This variation in the intensity has linear relation with the input electrical signal. Optical signal is then carried over by the optical fiber.

Receiver

There are various methods to configure detectors to extract digital data. Usually detectors are of linear nature. The detector SFH350V (Photo transistor) used in the kit has a transistor type output. The parameters usually considered in the case of detector are its responsivity at peak wavelength & response time. SFH350V (Photo transistor) has responsivity of about 0.8mA/10uW at 660nm.

But its response time is quite large & thus has lower bandwidth. When optical signal falls on the base of the phototransistor detector, proportional current flows through its emitter generating the voltage across the resistance connected between emitter & ground. This voltage is the duplication of the transmitted electrical signal, which is regulated using the Zener Diode to make it in the form of TTL output. Thus Detector is configured in the form of TTL detector. If the detected signal is lower than 5Volts output remains logic low (Zero) If the detected signal is equal to 5Volts output remains logic high (One).

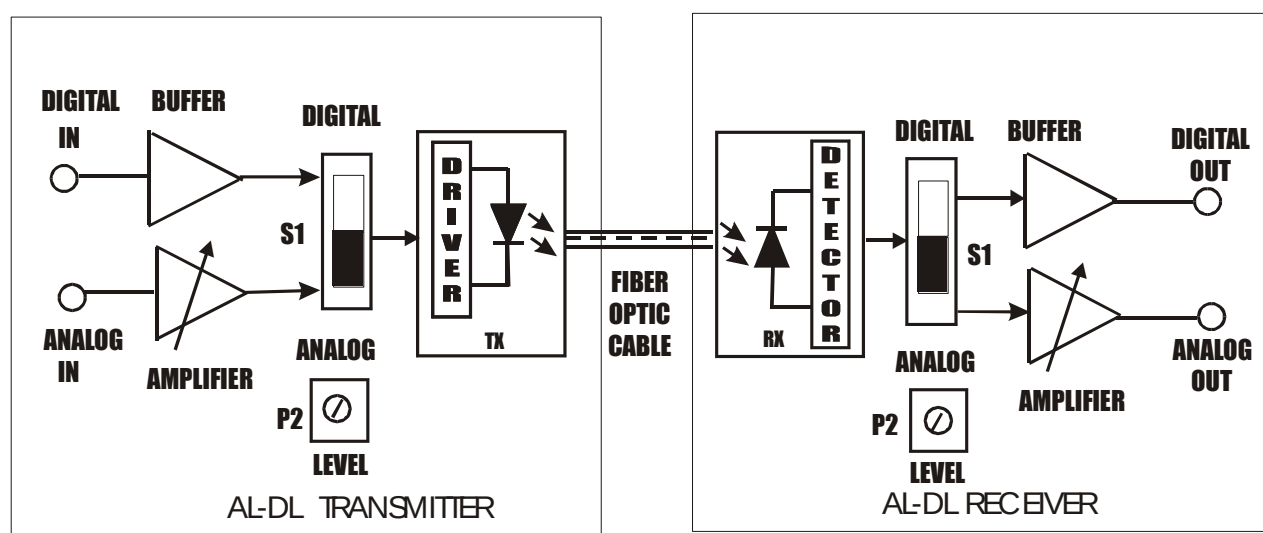
EQUIPMENTS

AL-DL TX & RX
Power supply
Dual Channel Oscilloscope
Function Generator
1meter Fiber Cable

PROCEDURE

- Slightly unscrew the cap of LED SFH 750V (660 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF. Now switch on the power supply.
- Put **S1** of TX and RX in **DIGITAL** position.
- Feed the TTL signal of about 1 KHz to **IN** post of Digital Buffer Section.
- Connect the one end of fiber to LED SFH 750V (660 nm) and other end to detector SFH 350V very carefully as per the instructions in step 1.
- Observe the received signal on CRO at **DIGITAL OUT** post.
- To measure the digital bandwidth of the link, vary the frequency of the input signal from 100Hz onwards and observe the effect on received signal.

EXPERIMENT NO.2



EXPERIMENT NO. 2

NAME

Setting Up a Fiber Optic Analog Link

OBJECTIVE

The objective of this experiment is to study a 660nm Fiber Optic Analog Link. In this experiment, you will study the relationship between the input signal & received signal.

THEORY

Fiber Optic Links can be used for transmission of digital as well as analog signals. Basically a fiber optic link contains three main elements, a transmitter, an optical fiber & a receiver. The transmitter module takes the input signal in electrical form & then transforms it into optical (light) energy containing the same information. The optical fiber is the medium, which carries this energy to the receiver. At the receiver, light is converted back into electrical form with the same pattern as originally fed to the transmitter.

Transmitter

Fiber optic transmitters are typically composed of a buffer, driver & optical source. The buffer electronics provides both an electrical connection & isolation between the transmitter & the electrical system supplying the data. The driver electronics provides electrical power to the optical source in a fashion that duplicates the pattern of data being fed to the transmitter. Finally the optical source (LED) converts the electrical current to light energy with the same pattern. The LED SFH750V supplied with the link operates within the visible light spectrum. Its optical output is centered at wavelength of 660 nm.

The LED **SFH750V** used in the link is coupled to the transistor driver in common emitter mode. The driver is preceded by the amplifier buffer. The amplifier in this case is a LF356 operational amplifier configured as voltage follower in the absence of input signal. In the absence of input signal half of the supply voltage appears at the base of the transistor. This biases the transistor near midpoint within the active region for linear applications. Thus LED emits constant intensity of light at this time. When the signal is applied to the amplifier it overrides the DC level at the base of the transistor, which causes the Q point of the transistor to oscillate about the midpoint. So the intensity of the LED varies about its previous constant value. This variation in the intensity has linear relation with the input electrical signal. Optical signal is then carried over by the optical fiber.

Receiver

The function of the receiver is to convert the optical energy into electrical form, which is then conditioned to reproduce the transmitted electrical signal in its original form. The detector SFH350V (Photo transistor) used in the kit has a transistor type output. The parameters usually considered in the case of detector are its responsivity at peak wavelength & response time. SFH350V (Photo transistor) has responsivity of about 0.8mA/10uW at 660nm. But its response time is quite large & thus has lower bandwidth. When optical signal falls on the base of the phototransistor detector, proportional current flows through its emitter generating the voltage across the resistance connected between emitter & ground. This voltage is the duplication of the transmitted electrical signal, which is amplified using the amplifier stage configured around Op-Amp IC LF356. Amplifier Gain control is done using Pot P2.

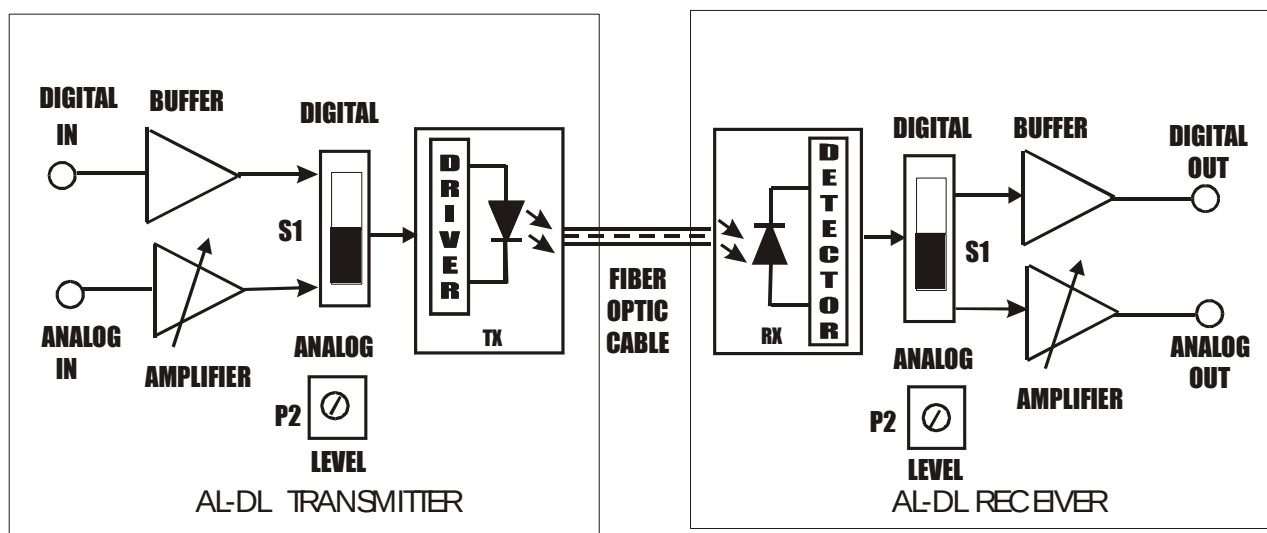
EQUIPMENTS

AL-DL TX & RX
Power supply
Dual Channel Oscilloscope
Function Generator
1meter Fiber Cable

PROCEDURE

- Slightly unscrew the cap of LED SFH 750V (660 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.
- Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF. Now switch on the power supply.
- Put **S1** of TX and RX in **ANALOG** position.
- Feed the ANALOG signal of about 1 KHz Sine wave, amplitude of about 1Vpp to **IN** post of Analog Amplifier Section.
- Set the gain of the amplifier of about two times higher to get 2V output at amplifier output section by pot P2.
- Connect the one end of fiber to LED SFH 750V (660 nm) and other end to detector SFH 350V very carefully as per the instructions in step 1.
- Observe the received signal on CRO at **ANALOG OUT** post. For adjustment of amplitude adjust P2 of RX Section.
- To measure the analog bandwidth of the link, connect the external Signal Generator with 1Vp-p sine wave to **IN** post of amplifier Section and vary the frequency of the input signal from 100Hz onwards and observe the effect on the output signal.

EXPERIMENT NO.3



EXPERIMENT NO.3

NAME

Study of Losses In Fiber Optics Analog Link

OBJECTIVE

The objective of this experiment is to measure propagation loss in plastic fiber provided with the Lab for different wavelengths of radiation as 950 nm, 660nm and also to measure the bending loss.

THEORY

Optical fibers are available in different variety of materials. These materials are usually selected by taking into account their absorption characteristics for different wavelengths of light. In case of optical fiber, since the signal is transmitted in the form of light, which is completely different in nature as that of electrons, one has to consider the interaction of matter with the radiation to study the losses in fiber. Losses are introduced in fiber due to various reasons. As light propagates from one end of fiber to another end, part of it is absorbed in the material exhibiting absorption loss. Also part of the light is reflected back or in some other directions from the impurity particles present in the material contributing to the loss of the signal at the other end of the fiber. In general terms, it is known as propagation loss. Plastic fibers have higher loss of the order of 180 dB/Km. Whenever the condition for angle of incidence of the incident light is violated the losses are introduced due to refraction of light. This occurs when fiber is subjected to bending. Lower the radius of curvature more is the loss. Another loss are due to the coupling of fiber at LED & photo detector ends.

EQUIPMENTS

AL-DL TX & RX
Power supply
Dual Trace Oscilloscope
1 & 3 Meter Fiber Cable

PROCEDURE

- Slightly unscrew the cap of LED SFH 750V (660 nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tight the cap by screwing it back.

- Connect the power supply cables with proper polarity to kit. While connecting this, ensure that the power supply is OFF. Now switch on the power supply.
- Feed the ANALOG signal of about 1 KHz Sine wave, amplitude of about 1Vpp to **IN** post of Amplifier Section.
- Set the gain of the amplifier of about two times higher to get 2V output at amplifier output section by pot P2.
- Connect the other end of Fiber to detector SFH 350V very carefully as per the instructions in step 1.
- Observe the received signal on CRO at **ANALOG OUT** post by adjusting **LEVEL** pot **P2** on Receiver section and you should get the reproduction of the original transmitted signal. Mark this amplitude as level V1.
- Now replace 1-meter fiber by 3-meter fiber without disturbing any of the previous settings. Measure the amplitude level at the receiver side again. You will notice that it is less than the previous one. Mark this as V2.
- If α is the attenuation of the fiber then we have,

$$P_1/P_2 = V_1/V_2 = e^{-\alpha(L_1 + L_2)}$$

Where

α = nepers / meter

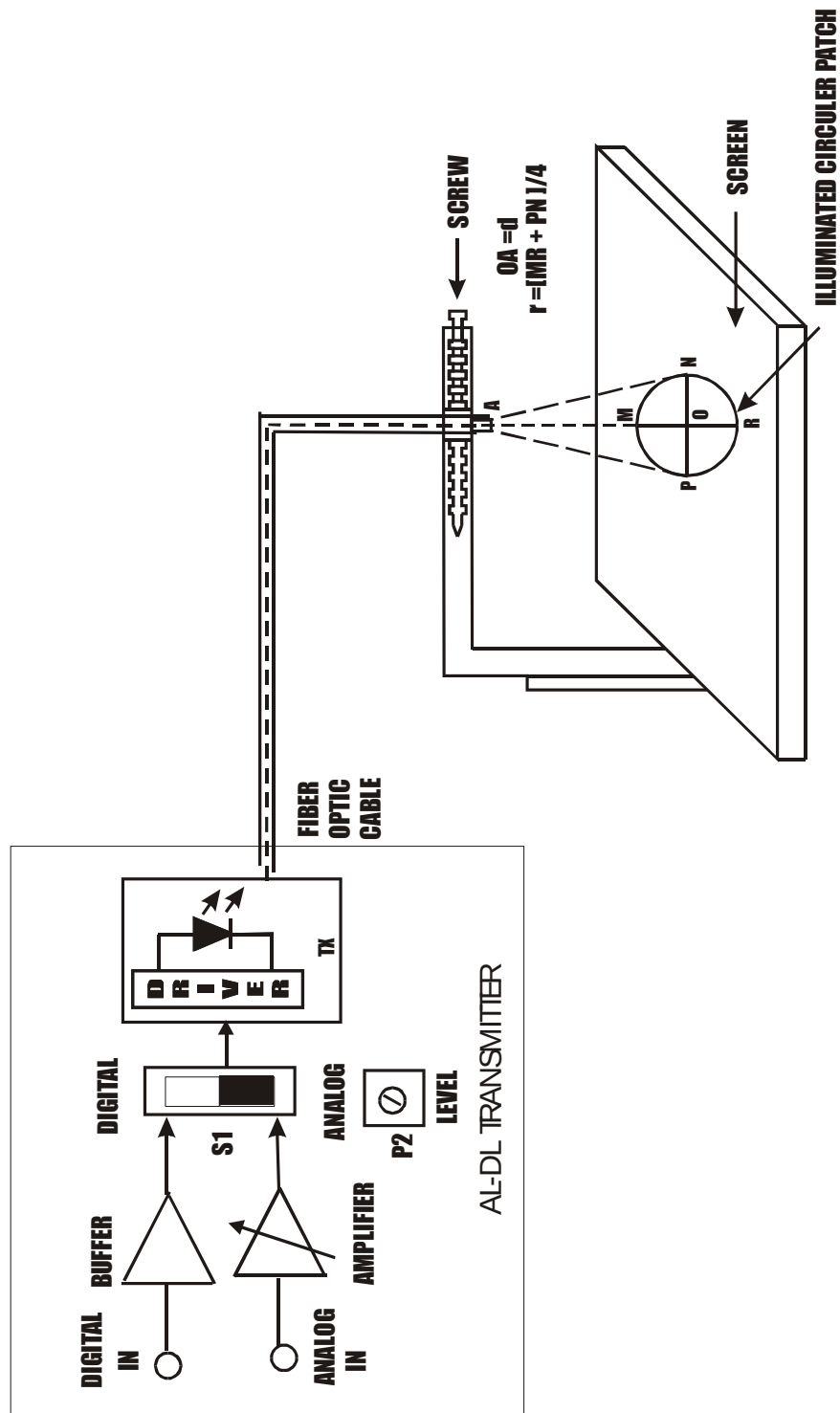
L_1 = fiber length for V_1

L_2 = fiber length for V_2

Measurement of Bending Losses

- Repeat all the Steps from. 1 to 4 as above.
- Bend the fiber in a loop. Measure the amplitude of the received signal.
- Keep reducing the diameter to about 1-2 cm & take corresponding output voltage readings. (Do not reduce loop diameter less than 1 cm.)
- Plot a graph of the received signal amplitude versus the loop diameter.

EXPERIMENT NO.4



EXPERIMENT NO. 4

NAME

Study of Numerical Aperture

OBJECTIVE

The objective of this experiment is to measure the numerical aperture of the plastic fiber provided with the kit using 660 nm wavelength LED.

THEORY

Numerical aperture refers to the maximum angle at which the light incident on the fiber end is totally internally reflected and is transmitted properly along the fiber. The cone formed by the rotation of this angle along the axis of the fiber is the cone of acceptance of the fiber. The light ray should strike the fiber end within its cone of acceptance, else it is refracted out of the fiber core.

Considerations in A Measurement

- It is very important that the optical source should be properly aligned with the cable & the distance from the launched point & the cable be properly selected to ensure that the maximum amount of optical power is transferred to the cable.
- This experiment is best performed in a less illuminated room.

EQUIPMENTS

AL-DL TX
Power supply
1-meter fiber cable
NA JIG
Ruler

PROCEDURE

- Slightly unscrew the cap of LED SFH750V (660nm). Do not remove the cap from the connector. Once the cap is loosened, insert the fiber into the cap. Now tighten the cap by screwing it back.
- Connect the power cord to the AL-DL TX & switch ON the power supply.
- Insert the other end of the fiber into the numerical aperture measurement jig. Adjust the fiber such that its cut face is perpendicular to the axis of the fiber.

- Keep the distance of about 10 mm between the fiber tip and the screen. Gently tighten the screw and thus fix the fiber in the place.
- Now observe the illuminated circular patch of light on the graph paper.
- Measure exactly the distance **d** and also the vertical and horizontal diameters **MR** and **PN** as indicated in the fig. 3.1.
- Mean radius is calculated using the following formula
 $r = (MR + PN) / 4$
- Find the numerical aperture of the fiber using the formula

$$NA = \sin \theta_{\max} = r / \sqrt{d^2 + r^2}$$

where $\theta_{\max}$ is the maximum angle at which the light incident is properly transmitted through the fiber.